APPENDIX E

1.0 Project Title. United States Botanic Garden (USBG) Elevator Preventive Maintenance Services.

1.1 Short Description. Contractor shall provide the USBG with elevator inspections, maintenance, and repair services. **In addition, the Contractor shall provide recertification and periodic testing to include: Category One Year Test, Category Three Year Test, and Category Five Year Test.** **NOTE:** All Category Tests shall be performed within the first year of the task order, and thereafter, shall follow a rigid testing schedule to be performed within three (3) months prior to the next testing expiration date.

1.2 Background. The Architect of the Capitol (AOC) is responsible to the United States Congress for maintenance, operation, development, and preservation of the facilities under its jurisdiction. The U.S. Botanic Garden (USBG) is a living plant museum that informs visitors about the importance, and often irreplaceable value, of plants to the well-being of humans and to earth's fragile ecosystems.

1.3 Objectives. The primary objective is to provide an effective preventive maintenance program, combined with scheduled inspection and testing, to ensure safety, performance, economy, and continued operations of all USBG elevators.

1.4 Equipment. The USBG has three (3) hydraulic elevators at two (2) separate facility locations:

1. Two (2) elevators (1 passenger and 1 freight) are located at the USBG Conservatory, 100 Maryland Avenue SW, Washington, DC.
2. One (1) freight elevator is located at the USBG Production Facility, 4700 Shepherd Parkway SW, Washington, DC.

Details for the three (3) hydraulic elevators located at two (2) locations are as follows:

Equipment at the USBG Conservatory				
Elevator Number	Manufacturer	Type	Capacity	Speed; Stops
# 1	Dover	Hydraulic Passenger	2,500 lbs	50 fpm; 2 stops
# 2	ThyssenKrupp	Hydraulic Freight	4,000 lbs	125 fpm; 2 stops

Equipment at the USBG Production Facility				
Elevator Number	Manufacturer	Type	Capacity	Speed; Stops
# 1	Motion Control	Hydraulic Freight	15,000 lbs	50 fpm; 2 stops

Please note that the passenger elevator at the Conservatory is located within the Tropics greenhouse. The conditions in this area are hot and humid all the time which increases the wear and tear of the elevator and components. In addition, stainless steel components have replaced the original equipment in several areas to extend the lifespan of those components.

2.0 Scope. All elevators shall conform to the elevator safety codes developed by the American Society of Mechanical Engineers (ASME). The USBG requires its elevators to be inspected and maintained in accordance with the current editions of ASME A17.1 - "Safety Code for Elevators and Escalators" **(the Code)** and ASME A17.2 - "Inspectors' Manual for Elevators and Escalators." In addition, USBG requires testing and certifying elevators in accordance with the Code. The contractor shall conduct the recertification testing to include an annual pressure test. This scope also includes repairs discovered during the course of providing preventive maintenance services.

2.1 General Maintenance. All maintenance, repairs, replacements, and alterations must conform to the applicable section of the Code. At no time, except when necessary during tests and inspections, may any required safety devices or electrical protection devices be made ineffective. In the event that these devices are made temporarily ineffective, they must be returned to full operating condition before the elevator can be returned to service.

2.2 Scope of Preventive Maintenance. The scope of the preventive maintenance program shall be comprehensive and include, at a minimum:

- Inspection
- Adjustments
- Lubrication
- Repairs and replacements (including light bulbs/lamps)
- Housekeeping/cleaning
- Recordkeeping

The Contractor shall maintain the elevator to run at rated speed, rated capacity, desired door open/close timing, designated floor stops, required floor leveling parameters, etc.

SEE SECTION A FOR DETAILED REQUIREMENTS

2.3 Maintenance Inspections. Specific items and equipment to be inspected under each of the following areas are listed in the Code and described in ASME A17.2.1 (for *electric* elevators) and ASME A17.2.2 (for *hydraulic* elevators).

Maintenance inspections shall include monthly preventive maintenance, repair and replacement of worn or defective components, lubrication, cleaning, and adjusting as required for operation as designed. Only parts and supplies recommended by the original equipment manufacturer shall be installed.

Contractor shall furnish all labor, materials, tools, equipment, transportation, and management to perform this work. **The preventive maintenance shall include inspecting, cleaning, adjusting, and lubricating the following (but not limited to) areas of each elevator.** The contractor shall provide a written report after preventive maintenance, which will include conditions found, actions taken, and recommended repairs and upgrades (elevator components and energy-reduction potential). The areas of inspection include:

a. Inside car – door reopening device, stop switches, operating and control devices, car floor/landing sill, lighting, car emergency signal, car door, door closing force, power opening/closing of doors, vision panels, car enclosure, emergency exit, ventilation, signage, rated load, platform area, data plate, emergency power, restricted door opening, car ride, door monitoring, stopping accuracy;

b. Machine room – access, head room lighting, receptacles, machine enclosure space, housekeeping, ventilation, fire suppression, pipes, wiring, ducts, guarding of equipment, numbering/labeling, disconnecting means, controller wiring/fuses/ grounding, static control, overhead beam, machines and machine brakes, motor- generators, regenerated power, alternating current (AC) drives, sheaves, rope fastenings, terminal stopping devices, slack rope devices, governor, safeties, data plate;

Hydraulic elevators require inspection of their unique additional equipment and systems such as: heating, hydraulic power unit, relief valves, control valve, tanks, flexible hoses/fittings, supply line, shutoff valve, hydraulic cylinder, fluid loss record, pressure switch, data plate, governor, recycling operation.

c. Top of car – stop switch, light, outlet, operating device, refuge space, counterweight clearance, sheaves, normal/final terminal stopping devices, broken rope/chain/tape switch, leveling devices, data plate, emergency exit, counterweight, counterweight buffer, counterweight safeties, floor numbering, hoistway construction, smoke control, pipes/wiring/ducts, windows/projections/recesses/ setbacks, clearances, multiple hoistways, traveling cables/junction boxes, door equipment, car frame, guide rails, guide rail alignment, guide rail fastenings, governor/traction/ compensation ropes, rope fastening devices;

Hydraulic elevators require inspection of their unique additional equipment and systems such as: terminal speed limiting devices, anti-creep limiting devices, speed test, suspension rope, governor rope releasing carrier, governor rope, wire rope fastening/hitch plate, slack rope device, traveling sheave, counterweight.

d. Outside the hoistway - platform guard, hoistway doors, vision panels, hoistway door locking devices, access, power closing of hoistway doors, sequence operation, enclosure, parking devices, emergency access, separate counterweight hoistway, standby power selection switch, emergency doors in blind hoistways;

e. Pit - access, lighting, stop switch, condition, clearance, runby, buffers, normal/final terminal stopping devices, traveling cables, governor rope, governor rope tension, compensating chains/ropes/sheaves, car frame/platform, car safeties, car guides;

Hydraulic elevators require inspection of their unique additional equipment/systems such as their plunger and cylinder.

f. Firefighters’ emergency operation- test for proper operation, including documentation;

g. Materials and Parts- provide all incidental materials, which includes rags, hydraulic oil, oil spill containment materials, lamps including LED type, non-control fuses, lubricant, cleaners, etc.

2.4 Twenty-Four (24) Hour Call Back Service. Entrapments or accidents involving vertical transportation equipment are an emergency call back, as is a condition where a unit is totally out of service. These calls shall have a **1 hour maximum ON-SITE response time**. Other call backs shall carry a maximum response time of 24 hours.

2.5 Testing. The Code requires periodic testing of elevators witnessed by a third party Qualified Elevator Inspector (as defined by ASME QEI-1). Test certifications may be issued by various public jurisdictions, such as State, county, or city elevator commissions or boards, or by private entities; however, in either instance, the 3rd party QEI shall be 100% independent and not an employee or have any ownership relationship with the Contractor.

The Contractor shall schedule and coordinate all inspections and testing—and notify the COR in writing at least thirty (30) days in advance of a scheduled inspection/test.

Metal test tags are required to be installed in the machine room for the Category 5 (full load) *electric* elevator test and for the Category 1, 3, and 5 *hydraulic* elevator tests.

SEE SECTION B FOR DETAILED REQUIREMENTS

2.5.1 Test Intervals. The interval for Category 1 tests is 12 months, for Category 3 tests is 36 months, and for Category 5 tests is 60 months. **NOTE: Category 1, Category 3, and Category 5 Tests are included in this SOW and all three (3) tests shall be performed within the first year of the task order, and thereafter, shall follow a rigid testing schedule to be performed within three (3) months prior to the next testing expiration date.**

For example:
Base Period (Year 1) – Perform Category 1, 3 & 5 Tests
Option Period 1 – Perform Category 1 Test
Option Period 2 – Perform Category 1 Test
Option Period 3 – Perform Category 1 Test & Category 3 Test
Option Period 4 – Perform Category 1 Test

2.5.2 Hydraulic Elevator Tests. The Code provides general descriptions of the Category 1, 3, and 5 tests for hydraulic elevators. More detailed descriptions are provided in ASME A17.2.2.

2.5.3 Category 1 Tests - Hydraulic Elevators. The Category 1 test requirements for *hydraulic* elevators involve the following equipment:

- Relief valve setting and system pressure
- Flexible hose and fitting
- Hydraulic cylinder leak test
- Standby power operation
- Firefighters' service
- Power operation of doors
- Normal and final terminal stopping devices
- Emergency terminal speed-limiting device
- Emergency terminal stopping device
- Pressure switch
- Oil buffer, Safety, Governor (if provided)
- Low oil test

2.5.4 Category 3 Tests - Hydraulic Elevators. The Category 3 test requirements for *hydraulic* elevators involve the following equipment:

- Unexposed portions of pistons
- Pressure vessels (hydrostatic test)

2.5.5 Category 5 Tests - Hydraulic Elevators. The Category 5 test requirements for *hydraulic* elevators involve the following equipment:

- Oil buffer (if provided)
- Safety (if provided)
- Governor (if provided)
- Coated ropes (if provided)
- Rope fastening on pistons (if provided)
- Overspeed valve

2.6 Government Inspections. The Government reserves the right at any time to have its own in-house elevator technicians or contract inspectors present during any work or inspections performed by the Contractor. The Government will have one or more competent elevator technician(s) or inspector(s) in attendance for all Category Tests.

3.0 Additional Requirements.

All travel, shipping, and parking expenses shall be borne by the Contractor performing this scope of work.

After each monthly preventive maintenance service is completed, within 10 working days, a written report including conditions of above systems found, actions taken, recommendations for repairs and upgrades, and energy-savings recommendations shall be provided to the USBG Capital Projects and Facilities Manager with a second copy provided to the Operations Supervisor (See *Section 4.0*). The report shall identify what was inspected, the status, and repair recommendations, if any.

*If additional work is required, provide written quotes on all additional repairs not mentioned in this scope of work. Any additional government task order(s) or a modification to the issued task order will be issued for additional repairs needed. Do **NOT** begin work until the additional task order or modification is issued **by an AOC Contracting Officer.***

The Contractor is to provide a written schedule of work to the USBG Capital Projects and Facilities Manager and the Operations Supervisor for approval *prior* to beginning any preventive or corrective maintenance work.

The Contractor is responsible for the recycling and disposal of all components removed in the course of this work in accordance with all applicable Federal, state, and local laws and regulations.

3.1 Place of Performance/Hours of Operation.

Monthly services shall occur during the 2nd full week of each month during normal work hours unless a different date is agreed to in writing in advance. The Contractor shall contact the COR <u>at least</u> 24 hours before a visit to either facility. The work for both locations shall be scheduled during normal working hours of 7:00 am to 4:00 pm (local time), Monday through Friday. Note that the USBG Conservatory is generally open to the public from 10:00 am to 5:00 pm, seven (7) days a week. Public areas will be blocked off by USBG staff as necessary.

Two (2) elevators (as described above) are located within the USBG Conservatory:
100 Maryland Avenue SW
Washington, District of Columbia 20024-3200

One (1) elevator (as described above) is located within the USBG Production Facility:
4700 Shepherd Parkway SW
Washington, District of Columbia 20032-5203

3.2 Period of Performance. The period of performance will consist of a 12-month Base Period and four (4) 12-month Option Periods for a total period of performance of five (5) years. FAR 52.217-8, Option to Extend Services is applicable to this scope of work, and thus, may extend the life of this requirement up to six (6) months.

3.3 Government Furnished Property/Equipment. There will be no government furnished property or equipment supplied for any resulting task order.

3.4 Contractor Furnished Items. The Contractor shall furnish all required equipment, tools, and materials to perform the services specified in this scope of work.

3.5 Maintenance Personnel. In accordance with the Code, maintenance, repairs, or replacements are to be performed only by persons trained to perform these operations on the equipment. The Contractor agrees to provide skilled, competent employees trained by the elevator manufacturer or an accredited elevator apprentice program for the purpose of maintenance and repair of elevators as defined in ASME 17.1, Safety Standard for Elevators and Escalators.

3.6 Key Personnel. Contractor shall provide the names of individuals that may be identified as key personnel immediately after issuance of the task order award and coordinate directly with the Government's representatives identified in *Section 4.0*.

3.7 Recordkeeping. The Contractor shall provide a written Maintenance Control Plan to include all units. A complete log must be maintained that contains records of all maintenance, adjustments, repairs, replacement, etc., performed on each elevator. The log must include the dates, names of participating personnel, and description of tasks performed, including tests and inspections, reports, trouble calls, corrective actions, recommendations, or any other incidents related to an elevator. All documents and service-related records shall be maintained on-site.

Manufacturer's data and drawings for the elevator equipment shall be accessible and maintained to reflect the current state of the equipment. Important data such as manufacturer names, part numbers, serial numbers, sizes, and types shall be readily accessible. Any pertinent service bulletins shall also be maintained on-site, either in the elevator mechanical room or the USBG Operations & Maintenance Supervisor's Office.

Checklists for the scheduled preventive maintenance tasks shall be developed and maintained on-site to ensure that these tasks are performed.

Any and all drawings, sketches, or other data generated in the performance of the task order will become the property of the AOC/USBG at the end of the task order. The format of this information or data shall be in customary formats (pdf, doc, xls, etc.).

3.8 Safety. The performance of this project will require personnel to work in public and areas occupied by USBG personnel. All necessary safety considerations will need to be taken. Areas will need to be blocked off, if necessary, to ensure the safety of visitors, staff, and others. As detailed above in the Scope, the Contractor is responsible for providing signs and utilizing them correctly to ensure the public and USBG staff are safe. The following practices shall be observed, at a minimum, during maintenance, inspection, or testing procedures:

- All safety devices must be in operational condition.
- Lockout/tagout procedures must be followed if maintenance procedures require that the equipment not be operated. The Contractor must lockout and tag-out all electrical circuits prior to beginning work. The Contractor is not permitted to work on energized electrical circuits without written approval from the Contracting Officer's Representative (COR). Qualified personnel must perform all electrical work. Schedule any power outages no less than 48 hours in advance.
- Ensure that personnel performing maintenance, inspection, and testing tasks wear clothing that is not loose fitting and that they are provided with proper protective equipment, such as safety shoes, hard hats, eye protection, and hand protection.
- Provide barriers and signage, where applicable, especially at hoistway doors.
- Upon completion of work, remove any jumper wires that were used.
- It is possible that the elevator pit may be designated a "Permit Required Confined Space." The additional required safe procedures must be attended to in these cases.
- Provide proper lighting.
- Determine that adequate refuge space exists above and below the car.
- Ensure the working area is clean and dry.
- Safety Data Sheets for all chemicals to be used throughout the entire process must be provided to the Government's Safety Specialist at least seven (7) days in advance.
- Provide, erect, maintain, and remove all guards, railings, fences, barricades, lights, gates, etc. required for the protection of all personnel and visitors of the USBG.
- The Contractor is responsible for providing and using the proper fall protection and signage to keep work areas safe for the public and USBG staff.
- Provide and maintain an adequate number of hand fire extinguishers at convenient locations. Avoid all accumulations of flammable debris by removing daily. Take all other precautions necessary to prevent fire or accidents. Closely supervise the storage of on-site materials.
- Comply with all applicable laws and ordinances, rules, regulations, and orders of any public authority having jurisdiction for the safety of persons or property to protect them from damage, injury, or loss. Erect and maintain, as required by conditions and progress of the work, all necessary safeguards for safety and protection, including posting of danger signs and other warnings against hazards. The Contractor shall be solely responsible for initiating, maintaining, and supervising all safety precautions, hazards, and programs (i.e., National Fire Protection Association standards) in connection with the scope of work.

4.0 Points of Contact.

The Contracting Officer for the resulting task order will be Ms. Tatjana Johnson, tatjana.johnson@aoc.gov, 202.262.0934.

The Contracting Officer's Representative (COR) for the resulting task order will be Mr. Patrick Andriuk, Capital Projects and Facilities Manager, patrick.andriuk@aoc.gov, 202-258-6217. The duties of the COR will not be assigned until the task order is awarded and issued by the Contracting Officer to the Contractor. This assignment will include all duties and limitations assigned to the COR.

At the Conservatory, the on-site POC will be Mr. Steve Weaver, Operations Supervisor, Stephan.Weaver@aoc.gov, Work: 202-226-4784 / Mobile: 202-483-8821 and shall be included in all scheduling and coordination.

At the Production Facility, the on-site POC will be Mr. Rudy West, Operations Assistant Supervisor, rudy.west@aoc.gov or 202-438-9872 and shall be included in all scheduling and coordination.

5.0 Applicable Policies and Standards.

a. **Americans with Disabilities Act - Accessibility Guidelines (28 CFR Part 36 Appendix A) (ADAAG).** The requirements of ADAAG include provisions that promote the safety and accessibility of disabled persons who use elevators. USBG elevators that are used by the general public or disabled facility personnel shall be considered for upgrade to meet ADAAG requirements.

b. **Code Data Plate.** ASME A17.1 requires that a data plate, indicating the Code edition in effect at the time of installation and the Code edition at the time of any alteration, be located in plain view and attached to either the main line disconnect or the controller. The Code edition(s) indicated on the data plate govern the testing and inspection requirements.

c. **Acceptance Inspections and Tests of New or Altered Installations.** ASME A17.1 requires that new or altered installations of both electric and hydraulic elevators be subjected to acceptance inspections and tests before being placed in service. The series of inspections and tests for new installations is comprehensive (inside car, machine room, top of car, outside hoistway, pit, and firefighters' emergency operation) and is detailed in the Code. The series of inspections and tests for altered installations generally involves those items that were affected and depends on the nature and effects of the alterations. These inspections and tests are also detailed in the Code.

SECTION A
MINIMUM PREVENTIVE MAINTENANCE
REQUIREMENTS FOR HYDRAULIC ELEVATORS

The following items (as applicable) shall be included requirements in the resulting maintenance task order. (Note: When items are inspected and found to be defective, broken, out of adjustment, etc., they must be repaired, replaced, or adjusted to meet the requirements of the current Code.)

<u>PERFORMED ON A **MONTHLY** BASIS</u>:

- Inspect and lubricate (as required) the machinery, pumps, piping, drive, valves, selector, and controller.
- Ride in the car and observe operation of doors, leveling, smoothness, and door reopening devices at each landing. Listen for unusual noises in the car and in the hoistway. If excessive creeping is occurring, determine cause and correct.
- Check all car operating controls, lamps, and gongs. Replace burned-out lamps.
- Clean:
  - Drip pans (check oil levels of associated equipment)
  - Door reopening device photo eye components
  - Door tracks and sills
  - Lamps and sensors in the car top controller
  - Car top
  - Machine room pit
  - Controller selector
  - Relay connectors contacts
- Inspect plunger seals and correct excess leakage.
- Fixtures, Indicators, and Buttons:
  - Buttons and Key switches: Shall have the correct legible markings. Must not stick or be plugged. Damaged buttons shall be replaced.
  - Indicators and Signals: Indicator lamps shall illuminate as required. The use of neon lamps, LED, or other long life light sources shall be encouraged. Broken lenses shall be replaced. Audible indicators shall function.
  - General: Face plates shall be in place and mounted square or plumb. Fastening screws shall be of the proper type and missing screws shall be replaced.
- Check and adjust the car door operation. Lubricate the hangers, rollers, gibs, linkages, and pivot points. Check and adjust door clearances, eccentrics, arm bearings, speed control switches, cables, clutches, chains, and belts. Tighten the door drive system points.
- Check the car telephone and alarm operations. Repair the alarm system if required.
- Check the operation of limit and safety switches in hoistway and in/on the car.
- Test mechanism: Observe for proper operation of motor and pump, oil lines, tank, controls, plunger, packing, etc. Check the oil tank level. Check the packing of valves and cylinder for leakage and tighten if necessary.
- Check the car emergency light.
- Check the car ventilation system and heater.
- Initiate the Phase I firefighter recall service and check for proper operation to a minimum of two (2) floors under Phase II service.

PERFORMED ON A **QUARTERLY** BASIS:

- Check the car for proper leveling operation. Adjust if required. Check and clean the door switch contacts.
- Check the door speed control switches.
- Check the condition of resistors and mounting assemblies.
- Check the car top and hoistway for loose covers, vanes, or components. Inspect the traveling cables for damage.
- Inspect the counterweight rope for wear and lubrication (if equipped). Clean the governor rope (if equipped).
- Inspect the rope hitches, fastenings, and shackles (if equipped).

PERFORMED ON A **SEMI-ANNUAL** BASIS:

Comprehensive inspection: See *Section 2.3* of this document.

PERFORMED ON AN **ANNUAL** BASIS:

- Conduct Category 1 tests (and Category 3 and 5 tests, if due). Inspect the condition of the flexible hoses and replace as needed. Pressure hoses are required to be tested, inspected, and replaced according to the Code.
- Test a sample of the hydraulic fluid for viscosity, color, contamination, foaming, and other pertinent properties specified by the equipment manufacturer. Drain and replace the fluid if the tests show it does not meet the requirements of the equipment manufacturer.
- Clean the guide rails with solvent.
- Test the car emergency light for required illumination.
- Vacuum the dust from controllers and relays.
- Clean the hoistway.

SECTION B
HYDRAULIC ELEVATOR
PERIODIC TEST CHARTS

ONE-year periodic tests for A17.1d (2000) and earlier editions Category "ONE" for A17.1-2000 and later editions		
Test Type	**Reference Rule A17.1d (2000) and earlier**	**Reference Section A17.1-2000 and later**
Relief Valve Setting and System	1005.2a	8.11.3.2.1
Flexible Hose and Fitting	1005.2d	8.11.3.2.4
Hydraulic Cylinder Leak Test	1005.2b	8.11.3.2.2
Emergency Terminal Speed Limiting Device	1005.2c(8)	8.11.3.2.3(h) Terminal Speed Reducing Device
Standby or Emergency Power Operation	1005.2c(6) [1002.2g]	8.11.3.2.3(f) [8.11.2.2.7]
Power Operation of Doors	1005.2c(7) [1002.2h]	8.11..3.2.3(g) [8.11.2.2.8]
Normal Terminal Stopping Devices	1005.2c(1) [1002.2e]	8.11.3.2.3(a) [8.11.2.2.5]
Slack Rope Device	1005.2f	8.11.3.1.3(z)
Firefighters Service	1005.2c(5) [1002.2f]	8.11.3.2.3(e) [8.11.2.2.6]
Pressure Switch	1005.2e	8.11.3.2.5
Oil Buffer (where provided)	1005.2c(4) [1002.2a]	8.11.3.2.3(d) [8.11.2.2.1]
Safety (where provided)	1005.2c(3) [1002.2b]	8.11.3.2.3(c) [8.11.2.2.2]
Governor (where provided)	1005.2c(2) [1002.2c]	8.11.3.2.3(b) [8.11.2.2.3]
Low Oil Protection Test	--	A17.1b-2003 8.11.3.2.3i

THREE-year periodic tests for A17.1d (2000) and earlier editions Category "THREE" for A17.1-2000 and later editions		
Test Type	**Reference Rule A17.1d (2000) and earlier**	**Reference Section A17.1-2000 and later**
Piston Rods	1005.3a	8.11.3.3.1
Pressure Vessels	1005.3b	8.11.3.3.2

FIVE-year periodic tests for A17.1d (2000) and earlier editions Category "FIVE" for A17.1-2000 and later editions		
Test Type	**Reference Rule A17.1d (2000) and earlier**	**Reference Section A17.1-2000 and later**
Safety (where provided)	1005.4(a)	8.11.3.4.1 [8.11.2.3.1]
Governor (where provided)	1005.4(a)	8.11.3.4.1 [8.11.2.3.2]
Oil Buffer (where provided)	1005.4(a)	8.11.3.4.1 [8.11.2.3.3]
Overspeed Valve	--	8.11.3.4.4
Coated Ropes	1005.4(b)	8.11.3.4.2
Rope Fastening on Pistons	1005.4(c)	8.11.3.4.3